Clinic Appointment Scheduler (Java)

A robust, console-based Java application designed to streamline clinic operations. This project implements the four pillars of Object-Oriented Programming (OOP) and utilizes **Apache POI** for database-less persistence via Microsoft Excel.

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🌟 Project Overview

The **Clinic Appointment Scheduler** automates administrative tasks including patient registration, doctor scheduling, and medical record management. Unlike traditional apps that require complex SQL setups, this system uses a `.xlsx` workbook as a flat-file database, making it portable and lightweight.

✨ Key Features

- **Smart Booking:** Real-time availability checking for doctors to prevent double-booking.
- **Medical Records:** Comprehensive tracking of patient diagnoses and prescriptions.
- **Data Persistence:** Automatic saving/loading of system state using Apache POI.
- **Unique Identifiers:** Automated UUID generation for patients, doctors, and appointments.

🏗 OOP Concepts Applied

- **Inheritance:** `Patient`, `Doctor`, and `Staff` extend the abstract `Person` class.
- **Encapsulation:** All entity fields are `private` with access controlled via `getters/setters`.

- **Abstraction:** The `Person` class and `ExcelManager` interface hide complex implementation details.
- **Polymorphism:** Method overriding of `displayInfo()` allows for dynamic behavior at runtime.

📂 System Architecture

```
src/main/java/com/clinic/
├── entities/           # Domain Models (Person, Patient, Doctor, etc.)
├── service/            # Business Logic (Clinic.java)
├── data/               # Data Access Layer (ExcelManager.java)
└── Main.java           # Entry Point & Menu Interface
```

🛠 Installation & Setup

Prerequisites

- **Java SDK 11** or higher.
- **Apache Maven** for dependency management.

Build Instructions

1. **Clone the repository:**

   ```
   git clone [https://github.com/your-username/clinic-scheduler.git](https://git
   cd clinic-scheduler
   ```

2. **Compile and Package:**

   ```
   mvn clean package
   ```

   This creates a "fat JAR" in the `target/` directory containing all dependencies.

🚀 Usage Guide

To run the application, use the following command (ensure your `lib/` folder contains the necessary Apache POI JARs if not using the Maven JAR):

```
java -cp "bin;lib/*" com.clinic.Main
```

Main Menu Options:

1. **Register Patient:** Input name and contact to receive a unique Patient ID.
2. **Register Doctor:** Set doctor name and specialization.
3. **Book Appointment:** Match Patient ID with Doctor ID for a specific date/time.
4. **View Medical History:** Retrieve all past records for a specific patient.

📸 Screenshots

(Add your project screenshots here)

Main Menu	Patient Registration

👥 Contributors

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